

NIFTY intraday volatility-risk-premium research

From audited option-chain data to a cost-aware, defined-risk shadow candidate

Instrument: NIFTY index options, single nearest-listed-expiry proxy

Research window: 1 Jan 2021–15 Jul 2026

Entry/exit: ATM±3, fixed 60 minutes (180-minute audit ceiling)

Capital experiment: Rs. 10,00,000; historical, no live allocation

Abstract

We ask whether a causal, normalized intraday variance-risk-premium (VRP) curve can time defined-risk NIFTY option structures after realistic execution and capital constraints. The foundation contains 43,018,677 one-minute option rows across 1,371 expected trading sessions, independently recomputed Black–Scholes implied volatility (IV), date-aware contract rules, India VIX/spot joins and a six-slot research SPAN join. A full unconditional coverage audit limits defensible fixed-contract research to ATM±3 and at most 180 minutes; the primary exit is 60 minutes. Provider IV is not accepted at face value, realized volatility (RV) uses only information available at entry, and IV/RV are converted to matched intraday variance before VRP is ranked within the intraday clock.

The original level/direction rule, 12 percentile-cross variants, structure/horizon grid, tail-reversal rules and a causal feature rescue all fail net of one-lot costs. A later, explicitly post-hoc upper-85 percentile upward crossing with a short iron fly, three causal deterioration gates and a frozen quality switch is historically positive: 86 executions from 132 signals, Rs. 153,013 gross, Rs. 44,029 costs and Rs. 108,983 net under a Rs. 10 lakh sizing experiment. The largest rupee drawdown is Rs. 7,982; the largest percentage drawdown is 0.762% in a different episode. However, the composite-score later-period Spearman bootstrap interval crosses zero; 2024, post-20-Nov-2024 and event-week results are weaker; and the same archive influenced selection. The correct decision is therefore **reject standalone intraday VRP timing and retain one p85 candidate only for frozen forward shadow observation**.

Research decision. The p85 candidate is approved for this submission and a forward-shadow protocol, *not* for live-capital deployment. Current live allocation is Rs. 0.

Research path in one table

Stage	Question	Main evidence	Decision
Foundation	Can the raw archive support causal fixed-contract labels?	100% session coverage; strict ATM±3/180m path coverage 99.09%	Proceed inside boundary
H1–H3	Does VRP alone pay through condors/flyes/verticals?	Every aggregate primary family negative after one-lot execution	Reject standalone rule
H4	Can a tail state plus causal regime gates clear costs?	One-lot mean gross Rs. 292, cost Rs. 240, net Rs. 52; 86 executed	Shadow candidate
Sizing	Can Rs. 10 lakh allocate without excessive margin/drawdown?	Avg 6.5 lots; avg margin 31.93%; max drawdown 0.76%	Feasible point, not frontier
Robustness	Is the positive result stable enough for capital?	Holdout score $\rho_s = 0.214$, bootstrap CI [−0.066, 0.484]	Live gate fails

This note reports the result that exists, including the failed hypotheses. All rupee P&L is based on historical date-aware exchange lot sizes and completed fixed-contract exits unless stated otherwise.

1. Data engineering: acquisition, cleaning, BSM, SPAN and gold

1.1 Reproducible three-layer archive

The Dhan modules call `/v2/charts/rollingoption` for the rolling option surface and `/v2/charts/intraday` for NIFTY spot, India VIX and available futures inputs; contract identities and expiry rules are reconciled to official exchange evidence. **Bronze** retains immutable provider bytes and request manifests. **Silver** performs source-specific parsing, timezone normalization, type enforcement, key construction and exception quarantine. The pre-BSM quality layer enriches same-session spot/VIX and date-effective rules, then blocks severe anomalies. **Gold** adds independent BSM outputs and SPAN references without rewriting upstream facts. Writes are atomic, resumable and hash-manifested; duplicate excess is quarantined. Thursday-to-Tuesday expiry migration and holiday rollback are effective-dated. Historical lots are keyed by contract expiry; the current lot is 65 (weekly expiries from 6-Jan-2026, monthly from 27-Jan-2026).

Gold specification		Count	Coverage	Interpretation
Rows / Parquet partitions		43,018,677 / 67	3.743 GB	CALL 21,509,933; PUT 21,508,744
Trading dates		1,371 of 1,371	100.000%	No missing or unexpected expected sessions
Time range (IST)	2021-01-01 09:15 to 2026-07-15 15:29			One-minute rolling surface
BSM-ready / blocked		42,970,270 / 48,407	99.8875% ready	Severe data failures never enter the solver
Spot matched / blocked		42,988,752 / 29,925	99.9304% matched	Backward same-session join; no future rows
VIX matched / unavailable / blocked	38,262,338 / 4,561,054 / 195,285		88.95% matched	Research uses contemporaneous ATM IV when unavailable
SPAN exact BOD contract matches		42,718,832	99.3030%	299,845 unmatched exact contracts retained
Integrity violations		0	all audited invariants	Null keys, negative volume/OI, OHLC, duplicates, future joins, BSM invariants

1.2 Independent Black–Scholes inversion

For call/put prices C, P , spot S , strike K , rate $r = 10\%$, dividend yield $q = 0$, and ACT/365 time T to verified 15:30 IST expiry,

$$d_1 = \frac{\ln(S/K) + (r - q + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}, \quad C = Se^{-qT}N(d_1) - Ke^{-rT}N(d_2), \quad P = Ke^{-rT}N(-d_2) - Se^{-qT}N(-d_1).$$

No-arbitrage bounds are checked before inversion. Newton is attempted first and Brent provides a bounded fallback: 28,196,221 and 5,085,391 successful rows respectively; only 48 solver failures remain. Repricing residuals are approximately 5.46×10^{-12} at the median, 6.26×10^{-9} at p95 and 1.05×10^{-7} maximum. Provider IV is retained only as a comparison field.

1.3 Timestamp-aware SPAN treatment – and its limit

The research release joins BOD/ID1/ID2/ID3 reference slots and selects the latest matched slot not later than entry (for the original 895 H1 trades: BOD 298, ID1 473, ID2 119, ID3 5). This is timestamp-aware within the retained reference schedule and gives entry-specific scenario arrays. It is **not proof of historical archive arrival**: retained publication timestamps are unproven, and the strict point-in-time release yields no usable historical SPAN matches. Margin is therefore a conservative timestamp-scheduled reconstruction, not a claim of observed broker margin at that instant.

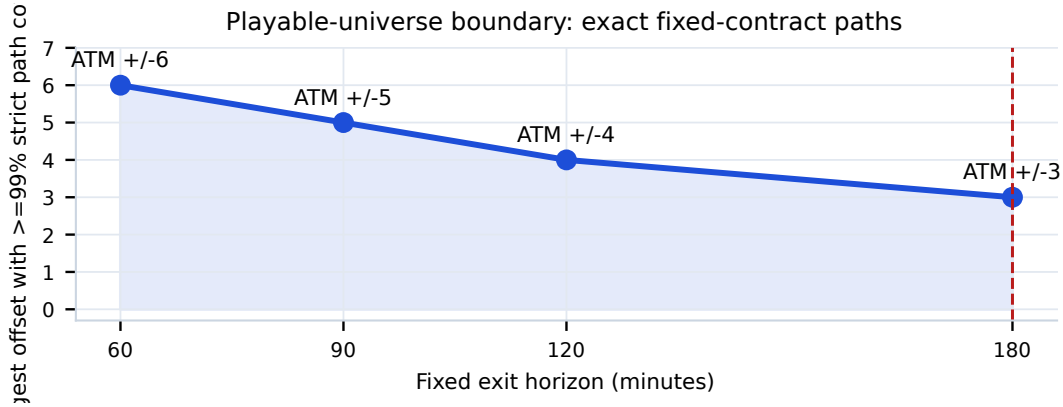
1.4 Dataset limitations carried into every conclusion

- The source is rolling ATM±10 WEEK history, not a complete absolute-strike chain. Evidence supports a *nearest-listed-expiry proxy*; it does not prove actual historical nearest-weekly identity in every row.
- No historical bid/ask, queue position or expired-futures minute archive exists. Fills require a conservative model and longer paths are non-randomly censored as ATM migrates.
- The versioned 751-row, three-date sample is for schema/reproducibility; full Parquets stay outside Git and are pinned by corpus SHA-256 `ae5db...252`.

2. Playable universe and volatility normalization

2.1 Unconditional moneyness $\times$ horizon audit

Every theoretical entry minute on every available session was audited at 15/30/60/90/120/180/240/300 minutes and offsets $\text{ATM} \pm 0$ through $\text{ATM} \pm 10$. Contract identity is frozen at entry; a later rolling label is never substituted. For the four-leg $\text{ATM} \pm 3$ window, 60-minute exact endpoints cover 99.7766% of 430,665 windows, strict minute paths cover 99.6754%, and a last-observation-carried-forward rule capped at ten minutes reaches 99.8546% with only 258 proxies (0.0599%). At 180 minutes, 266,698 windows have 99.5208% exact endpoints, 99.0926% strict paths and 99.7068% ten-minute-stale coverage; 621 proxies (0.2328%) remain.



The 99% strict-path frontier contracts from $\text{ATM} \pm 6$ at 60 minutes to $\text{ATM} \pm 3$ at 180 minutes. Entries begin at 09:16 (09:15 has a large source hole) and must leave an observable fixed exit before 15:29. Missingness is tail-correlated with ATM migration, so >180 -minute or multi-day profitability cannot be inferred from survivors.

2.2 IV, RV and VRP are matched to the intraday clock

The signal does not subtract an arbitrary daily RV statistic from a chain IV percentage, and it does not use the upstream gold spot-BSM IV at face value. Synchronized $\text{ATM}-1/\text{ATM}/\text{ATM}+1$ calls and puts imply a parity forward,

$$F_t = \text{median}_K \left\{ K + e^{rT} [C_t(K) - P_t(K)] \right\},$$

then CE and PE IV are independently inverted with discounted Black-76 and averaged at ATM. Local $\text{ATM} \pm 3$ CE/PE IV supplies skew and curvature; provider IV is reconciliation-only. The ATM variance forecast is $V_{t,H}^{IV} = \hat{\sigma}_{IV,t}^2 \tau_H$, where $\tau_H = H/(365 \cdot 24 \cdot 60)$. RV is built causally from one-minute log returns observed up to t . On a matching interval,

$$V_{t,H}^{RV} = \sum_{i=t-H+1}^t r_i^2, \quad \hat{\sigma}_{RV,t,H} = \sqrt{V_{t,H}^{RV} / \tau_H}, \quad VRP_{t,H} = V_{t,H}^{IV} - V_{t,H}^{RV}.$$

The implementation also retains annualized variance spread $\hat{\sigma}_{IV}^2 - \hat{\sigma}_{RV}^2$, $\log(IV/RV)$, IV–RV divergence and 5/15/30-minute changes. Matching conventions makes the magnitude economically interpretable; common positive scaling preserves crossing direction.

2.3 Causal curve level, direction and acceleration

Intraday IV/RV are strongly seasonal. We therefore rank VRP against historical observations from the same minute-of-day, using only prior dates:

$$q_t = \frac{1 + \sum_{s < t, m(s)=m(t)} \mathbf{1}[VRP_s \leq VRP_t]}{n_{t,m} + 1}, \quad v_t = q_t - q_{t-5}, \quad a_t = v_t - v_{t-5}.$$

The hypothesis family uses zero crossings, percentile-tail crossings, v_t , a_t and tail reversals. Signals execute on the next exact minute; exit is the same frozen contracts at $t + 60$ (120/180 only as declared diagnostics). One signal per session avoids overlapping duplicates in the base tests.

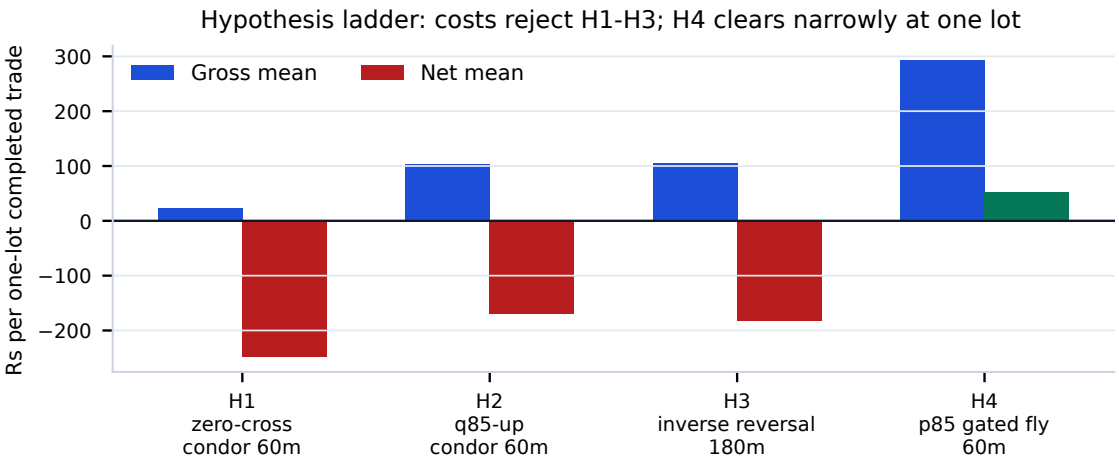
2.4 Initial number crunching

Unconditional matched 60-minute VRP is positive in 39.54% of windows. Complete $\text{ATM}/-3\text{PE}/+3\text{CE}$ surfaces exist in 511,859 windows (99.88%). Raw rank dependence is weak: Spearman correlations of short-condor P&L with VRP percentile, percentile velocity, percentile acceleration and raw-VRP acceleration are respectively 0.1381, 0.0147, -0.0012 and 0.0031. This motivates a falsifiable crossing test, but not a trading claim.

3. Hypotheses H0–H4: the full path, including failures

Naming. H_0 is the statistical null; H1–H4 are sequential trading alternatives, fixed or explicitly labeled post-hoc. The primary outcome is mean *net* rupee P&L per completed historical lot. A result must also survive tail, calendar, concentration and capital checks.

ID	Falsifiable statement	Implementation and result	Status
H_0	VRP curve level/direction has no positive cost-inclusive predictive value for 60-minute defined-risk P&L.	Rejection requires positive net mean and robust resampling/calendar evidence, not gross association alone.	Not rejected by H1–H3.
H_1	First sustained VRP crossing from non-positive to positive favors an ATM±3 short iron condor.	895 trades: gross mean Rs. 22.60, cost Rs. 271.14, net mean –Rs. 248.54; 9.39% net wins; 0/65 positive months.	Reject.
H_2	Deeper causal percentile crossings plus direction/velocity/acceleration improve short/long defined-risk timing.	70/75/80/85/90/95 thresholds × up/down: all 12 short-condor variants have negative net means. q85-up is least bad at the broad sample: 272 trades, gross Rs. 102.73, net –Rs. 169.32.	Reject.
H_3	Reversal from upper tail toward zero favors long condors; lower-tail reversal favors short condors (inverse pre-declared).	Requested 60m map: 428 trades, gross –Rs. 2.71, net –Rs. 294.12. Inverse 180m: gross Rs. 104.60 but net –Rs. 181.43 with 49.19% coverage.	Reject.
H_4	At an upper-85 upward crossing, a short iron fly becomes viable when causal IV/RV deterioration regimes are excluded and size is cost-aware.	132 signals; 86 pass gates. Equal one lot: gross Rs. 292.42, cost Rs. 240.36, net Rs. 52.06. Frozen capital path: Rs. 108,983 net.	Shadow only.



3.1 Why the condor barely moved while costs looked large

At 6–20 DTE, 60 minutes is a short window for a four-leg, partially Greek-offset condor: wing and body moves cancel, net vega/theta are modest, and the signal is not a direct tradable on the scalar VRP. Yet entry plus exit require eight orders. Under H1, the average bill was Rs. 160 brokerage, Rs. 59.18 modeled slippage, Rs. 30.39 GST, Rs. 12.37 STT, Rs. 8.68 exchange charges and about Rs. 0.53 other regulatory charges. Thus small gross structure movement, not an accounting bug, was the central failure.

3.2 Structure/horizon and longer-hold diagnostics

The cost-aware grid compared condors, iron flies, butterflies and two-leg verticals at 60/120/180 minutes: 48 cells, zero positive-net-mean cells. The best was a lower-10-down 180-minute bull call spread (219 trades, Rs. 139.16 gross, –Rs. 19.22 net). Some 180-minute gross means exceeded Rs. 65, consistent with a slower maturation story, but round-trip costs remained roughly Rs. 249–311. Multi-day testing produced only one cell with at least 80% coverage and it failed. For the apparent top-reversal 180-minute subgroup, 96 of 191 signals are evaluable and **95 are explicitly unevaluated**; the missing half cannot be assumed random.

3.3 Final causal feature rescue before closing standalone VRP

A feature model used IV/RV divergence, $\log(IV/RV)$, VRP level/slope/acceleration, local skew/curvature, spot returns, DTE, Greeks, credit/debit and cost hurdles. Its 2025–26 confirmation had 190 trades, gross –Rs. 6,130, net –Rs. 31,355, mean net –Rs. 165.02 and failed six of eight gates. Module 3 therefore closes the standalone VRP hypothesis family rather than mining another crossing.

4. H4: the one retained p85 regime-conditional candidate

4.1 Frozen trade contract

At the first causal upper-85 percentile crossing upward, enter next minute in a nearest-listed-expiry-proxy **short iron fly** with every leg inside $\text{ATM} \pm 3$ at entry; close the same contracts after 60 minutes. The structure is fully defined-risk. Historical lot size and latest research SPAN slot not later than entry are used. India VIX informs the slippage regime when present; contemporaneous reconstructed ATM IV $\times 100$ substitutes where absent.

Three discovery-period deterioration conditions reject entries:

$$\Delta IV_{5m} \leq -0.046817, \quad \Delta IV_{15m} \leq -0.114873, \quad \Delta \hat{\sigma}_{RV,5m} \leq -0.02651714.$$

The intuition is structure-specific: a short fly entered while IV/RV are already collapsing has too little remaining premium response relative to eight-order friction. Thresholds were learned on 2021–23 and frozen for later diagnostics; they are not a pristine prospective discovery.

4.2 Confidence score, rank test and hard quality switch

The causal score is 50% gate cushion, 15% inverse IV percentile, 15% inverse RV percentile, 10% inverse DTE percentile and 10% entry-time score. Percentile maps are frozen from 2021–23. Spearman is computed against *one-lot net P&L before sizing*:

$$\rho_s = \text{corr} \left(\text{rank}(\text{score}_t), \text{rank}(PnL_{t,1\text{lot}}^{\text{net}}) \right),$$

which prevents an artificial correlation from assigning more lots to high scores. Full-sample $\rho_s = 0.271$ (one-sided permutation $p = 0.0014$); discovery $\rho_s = 0.337$ with bootstrap CI $[0.100, 0.551]$; combined 2024–26 $\rho_s = 0.214$, one-sided $p = 0.0472$, but bootstrap CI $[-0.066, 0.484]$. Because the later lower bound is not positive, confidence-proportional sizing fails its strict validation gate. The final rule uses a simpler **hard switch**: execute only if score > 0.40 ; power zero means score does not scale lots after admission.

4.3 Cost model and corrected quantity impact

For integer lots L ,

$$PnL_t^{\text{net}}(L) = L PnL_{t,1}^{\text{gross}} - C_t^{\text{Groww}}(L) - C_t^{\text{slip}}(L).$$

Groww charges include date-aware sell-side STT, brokerage, exchange transaction charge, SEBI fee, stamp, IPFT and GST. Selected entries have at least 6.08 DTE and exit after 60 minutes, so expiry-settlement STT, expiry-day ELM and disappearing calendar-spread benefit are outside the realized trade path rather than omitted applicable costs. Synthetic adverse slippage begins with premium/VIX, depth, staleness and observed volume/OI terms. The removed 10%-per-extra-lot ladder was demonstrably dominant despite sub-1% participation. Its corrected incremental factor is

$$\frac{\text{impact}}{\text{base}} = \frac{L-1}{59} + \text{sqr}t \frac{q_{\text{inc}}}{V_{1m}} + \sqrt{\frac{q_{\text{inc}}}{OI}},$$

so the smooth heuristic contributes one base-slippage unit around 60 lots while participation retains the liquidity penalty. At six lots, the original model's added impact was 94.7% arbitrary ladder; the correction restores market variables as the main scaling logic. This is calibrated conservatism, not measured order-book impact.

4.4 Why H4 clears where the original condor does not

Across 86 admitted signals at one equal lot, gross P&L is Rs. 25,148, charges Rs. 17,270, slippage Rs. 3,401 and net Rs. 4,477: Rs. 292.42 gross, Rs. 240.36 total cost and Rs. 52.06 net per trade. The cushion is narrow at one lot. With exact historical profile sizing, gross averages Rs. 1,779 and total costs Rs. 512 per executed trade, leaving Rs. 1,267 net; fixed brokerage is amortized while the corrected impact curve remains moderate at 6.5 average lots. This is the specific economically viable candidate – not evidence that arbitrary VRP tails work.

Research decision. H4 is retained because gross clears modeled cost at one lot and by a wider amount under feasible sizing. It remains post-hoc and conditional; no claim is made that p85 is a universal optimum.

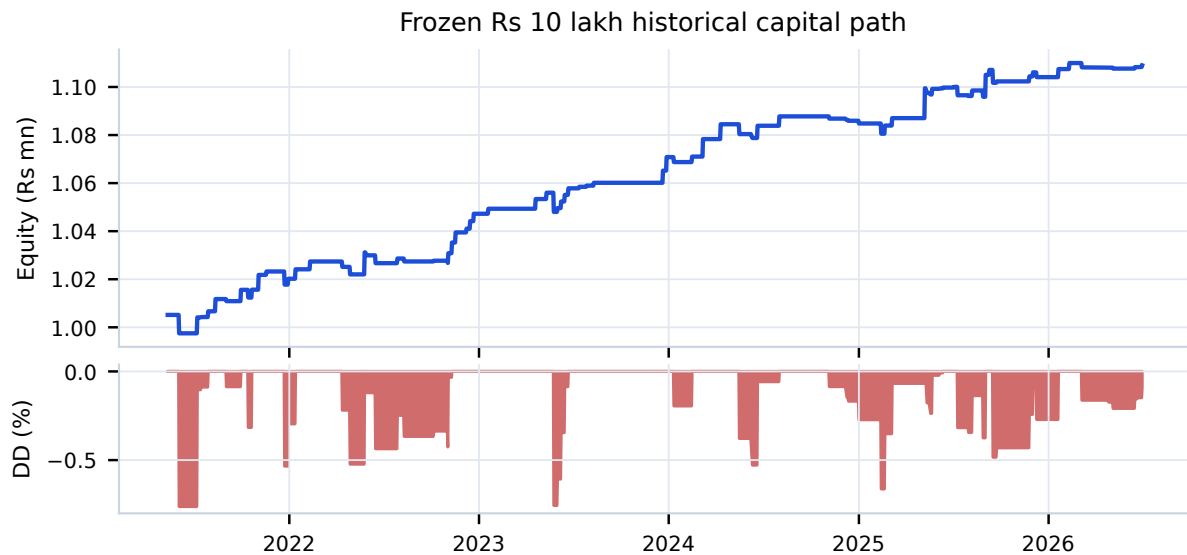
5. Rs. 10 lakh sizing, risk management and tearsheet

5.1 Sizing rule selected from a broad exploratory grid

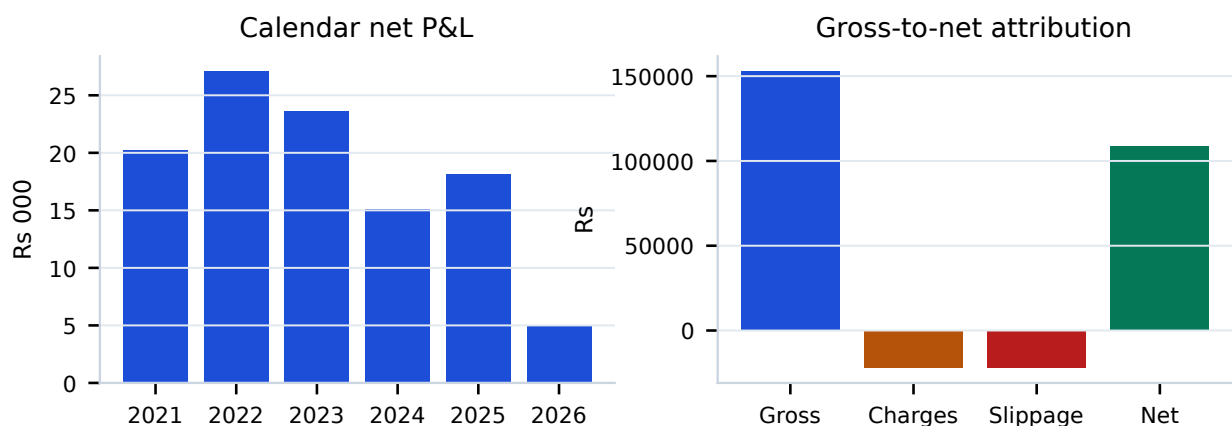
Starting equity is Rs. 10,00,000. For each admitted trade,

$$L_t = \min \left(\left\lfloor \frac{0.35E_t}{M_{t,1}} \right\rfloor, \left\lfloor \frac{0.04E_t}{Loss_{t,1}^{defined} + Reserve_{cost,1}^{q95}} \right\rfloor, 76 \right),$$

with integer lots and exact quantity-aware cost recomputation. The rule caps entry SPAN at 35% of current equity, defined max loss plus discovery q95 round-trip cost reserve at 4%, and capacity at 76 lots. There is no fitted drawdown or losing-streak brake. The 17,640-policy grid produced 4,392 discovery-eligible rules; discovery-vs-holdout policy-net rank correlation was 0.649. The chosen configuration (margin 35%, risk 4%, score floor 0.40, score power 0) lies in a useful neighborhood, but **is not a proven efficient frontier**: objectives, intratrade MTM paths, SPAN expansion and truly independent data are insufficient to identify a global risk-return optimum.



Metric	Result	Metric	Result
Candidate / executed / skipped	132 / 86 / 46	Average / maximum lots	6.50 / 14
Gross / costs / net	Rs. 153,013 / Rs. 44,029 / Rs. 108,983	Ending equity / total return	Rs. 1,108,983 / 10.90%
CAGR / annualized vol	2.04% / 1.43%	Sharpe / Sortino	1.37 / 2.75
Hit rate / profit factor	66.28% / 2.537	Avg win / avg loss	Rs. 3,156 / -Rs. 2,445
Worst trade/day/week	-Rs. 7,982	Trade CVaR 5%	-Rs. 6,142
Max rupee DD / max % DD	-Rs. 7,982 / -0.762% (different episodes)	Recovery / turnover / cost drag	184d / Rs. 6.67m / 28.77%
Average / max margin use	31.93% / 34.97%	Average / max risk use	2.79% / 3.98%

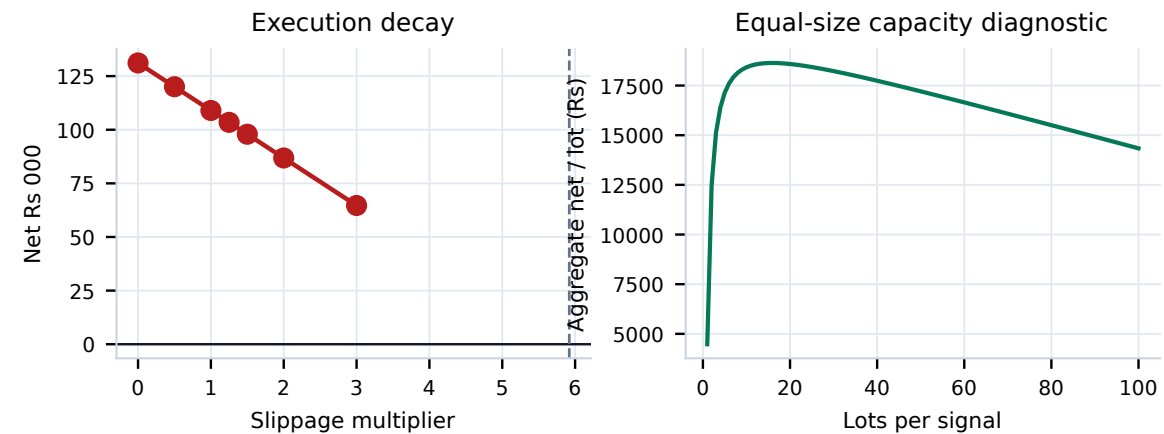


5.2 Calendar path and distribution

Net P&L is positive in every displayed calendar year: 2021 Rs. 20,204; 2022 Rs. 27,046; 2023 Rs. 23,585; 2024 Rs. 15,113; 2025 Rs. 18,148; 2026 through 14 July Rs. 4,888. Discovery 2021–23 contributes Rs. 70,835; 2024 validation Rs. 15,113; 2025–26 confirmation Rs. 23,036. Cumulative net divided by average entry margin is 32.11% (not annualized). The worst trade, 25-May-2023, was seven lots: -Rs. 7,577.50 gross plus Rs. 404.23 cost, -Rs. 7,981.73 net; it bound on reserved cash risk and recovered by 22-Jun-2023. This breadth is encouraging, but sparse dependence and the post-hoc path prevent clean-OOS language. Sharpe should not dominate judgement on a short-gamma structure without intratrade MTM tails.

6. Robustness: costs, capacity, regimes, events and failure modes

6.1 Execution decay and capacity



At the selected historical sizes, modeled aggregate P&L breaks even only near $5.92\times$ baseline slippage. The equal-size 1–100 lot diagnostic remains positive over the tested grid, but this is a model sensitivity, not proof of fillable exchange capacity. On Rs. 10 lakh, margin/risk caps bind far before 100 lots. Capacity outside observed minute volume/OI, concurrent orders and queue priority is unknown.

6.2 Regime, structural-break and event evidence



Pre-20-Nov-2024 has 53 trades and mean sized net Rs. 1,639; post-break has 33 and Rs. 671. The post-minus-pre per-selected-lot difference is $-\text{Rs. } 71.76$, bootstrap 95% CI $[-\text{Rs. } 377.81, \text{Rs. } 262.03]$, permutation $p = 0.639$: weaker, but not statistically distinguishable in this sample. Event windows use ± 5 business days around RBI MPC, Budget and national-election anchors: 27 event-week trades average Rs. 418 with 53.6% cost drag, versus 59 non-event trades at Rs. 1,656 and 24.1% drag. Heavy-results weeks are not tagged because a defensible index-specific calendar was unavailable. These event numbers are descriptive, not causal.

The exploratory losing-trade audit found useful *directions*, not validated filters: wide gate cushion, lower IV/RV, lower DTE and entries after 13:00 were better; high-IV/high-RV, thin-cushion and 11:00–13:00 groups were worse. The three deterioration gates were learned only on 2021–23 and kept fixed. Adding more regime cuts now would be another same-sample search.

6.3 What passed, what failed

Test	Outcome	Reading
One-lot H4 gross > charges + modeled slippage	Pass	Rs. 52.06 mean net, but a thin base-size cushion
Later-period score Spearman ≥ 0.20 and $p \leq 0.05$	Pass	$\rho_s = 0.214$, one-sided $p = 0.0472$
Later-period bootstrap lower bound > 0	Fail	95% interval $[-0.066, 0.484]$
Positive chronological capital slices	Pass	Discovery, 2024 and 2025–26 positive under frozen capital rule
November 2024 break stability	Inconclusive	Point estimate weaker; wide interval, small post sample
Multi-day / > 180m verification	Not evaluable	Rolling fixed-contract coverage censors outcomes
Intraday liquidation/tail path	Not observed	Equity records fixed-horizon completed trades, not worst MTM

6.4 Why this might be fake

The retained candidate was found after H1–H3 failed; p85, structure, gates and sizing all reuse a finite archive. No historical spread/queue data validates modeled fills. SPAN timing is a research schedule, not proven publication time. The expiry identity is a nearest-listed proxy. Missing longer paths are state-dependent. Score rank uncertainty spans zero, 2024 is weak, and event/post-break performance is lower. The strongest alternative explanation is post-selection plus fixed-cost amortization, not stationary regime alpha.

7. Decision, kill-switch, task coverage and reproducibility

7.1 Capital decision and frozen forward protocol

Research decision. Allocate Rs.0 live. Run the exact H4 rule in shadow at one modeled lot until new observations – not re-optimization – decide whether the candidate survives.

Minimum evidence before capital discussion: at least 100 new non-overlapping trades, at least 12 months and eight active months; positive net P&L overall and in both chronological six-month halves; no month above 40% of total forward profit; score/P&L Spearman bootstrap 95% lower bound above zero; modeled-vs-observed 20-fill slippage ratio no more than 1.5× and no fill above 3×; forward maximum drawdown below 1.5%. A future micro-allocation would start at one lot with a 0.75% daily capital stop, immediate data/timestamp/instrument kill-switch and no discretionary averaging. Any parameter change restarts the forward clock.

7.2 Quant-task requirements – explicit coverage

Brief requirement	Status	Evidence in submission
Specific falsifiable VRP hypothesis	Complete	H0–H4 statements, crossings and acceptance gates
Defined-risk NIFTY structure	Complete	Condors/flies/verticals only; H4 short iron fly
Data source, cleaning and limitations	Complete	Bronze/silver/gold, 43.0m-row audit, proxy/coverage disclosures
No lookahead / expiry mechanics	Complete	Prior-date minute ranks, next-minute entry, fixed contracts, dated rules
Realistic costs, slippage and margin	Complete	Groww charges, dated STT, quantity impact, entry SPAN; no mid fills
IS/OOS check and self-attack	Qualified	Chronological discovery/later slices; explicitly not pristine OOS
CAGR, vol, Sharpe, Sortino	Complete	Net-capital metrics with short-gamma caveat
Drawdown/duration, CVaR, worst day/week	Complete	Trade sheet and risk artifacts; intratrade MTM limitation
Hit rate, win/loss, turnover, cost drag	Complete	Page 6 tearsheet and formula-backed workbook
Regime and Nov-2024 break	Complete	Causal gates, formal break/permutation/bootstrap diagnostics
Events and execution decay	Complete	RBI/Budget/election windows; 0–6× slippage curve
Capacity and solo-desk monitoring	Complete	Quantity-aware 1–100 lot diagnostic; shadow/kill-switch design
Trade/size/kill decision	Complete	Shadow only, Rs.0 live; exact promotion conditions

7.3 Reproduce and inspect

From the repository root, run:

```
.\research\module5_final_submission\scripts\run_submission.ps1
```

This rebuilds compact analytics, verifies hashes, regenerates this LaTeX PDF and formula-backed workbook, and runs the test suite. The reviewer surface includes `submission/NIFTY_VRP_Research_Memo.tex/.pdf`, `NIFTY_VRP_Research_Tearsheet.xlsx`, all 132 signal rows with 46 explicit skips, equity/monthly/weekly/annual curves, drawdown episodes and autopsy, charge/slippage attribution, capacity/decay tables, break/event diagnostics, manifests and module runbooks. Raw multi-GB Parquets remain outside Git by design; full-corpus SHA, audited sample and routing contract pin the source.

7.4 Conclusion

The central result is a disciplined rejection: normalized intraday VRP level, tails, direction, velocity, acceleration and reversal do not by themselves generate enough 60–180 minute defined-risk movement to cover execution. A p85 short-iron-fly regime candidate produces a meaningful historical gross-to-cost cushion after gating and feasible sizing, with broad positive calendar P&L and controlled completed-trade drawdown. But its discovery is post-hoc, rank stability is not decisive and market/data boundaries are material. The strategy is therefore complete enough to submit, not complete enough to fund.

Primary references

NSE, [Equity derivatives contract specifications](#); NSE circulars [FAOP68685](#) (Tuesday expiry) and [FAOP70616](#) (lot 75 to 65); SEBI, [Measures to strengthen index derivatives](#) (Nov-2024 regime); [RBI MPC archive](#); [Union Budget](#); [Election Commission 2024](#). Internal evidence: gold audit, Phase 2 universe/volatility notes, Module 3 hypothesis closeout, Module 4 sizing closeout and Module 5 final manifest.

Submission status: exact strategy/config frozen; shadow-only; no live-capital approval.